$\lvert N \rvert$	Model	$\mathrm{RI}(\mathcal{M}^{\text{late}}_{-1})$	$\mathrm{RI}(\mathcal{M}^{\text{late}}_{-1})$ **Result Fixed**
2	GPT-J	40.2%	4.4%
	Pythia 2.8B	43.2%	5.8%
	LLaMA 7B	36.1%	7.5%
	Goat	33.5%	7.4%
	GPT-J (Words)	27.8%	4.5%
3	Pythia 2.8B	13.5%	6.7%
	Pythia 2.8B (FT)	24.7%	13.6%

Table 1: Relative importance (RI) measurements for the last-token late MLP activation site. The decrease in the RI observed when fixing the result of the two pairs of operands used for the interventions quantitatively confirms the role of this subset of the model in incorporating result-related information.

where the subscript -1 indicates the last token of the input sequence and L is the number of layers in the model. This quantity represents the relative contribution of the mid-late last-token MLPs compared to all the MLP blocks in the model.

For the two-operand setting, we carry out the experimental procedure described in Section 3 for three additional models: Pythia 2.8B, LLaMA 7B, and Goat.[5] Furthermore, we repeat the analyses on GPT-J using a different number representation: instead of Arabic numbers (e.g., the token *2*), we represent quantities using numeral words (e.g., the token *two*). For the three-operand setting, we report the results for Pythia 2.8B before and after fine-tuning. We measure the effects using both randomly sampled and result-preserving operand pairs, comparing the RI measure in the two settings. The results (Table 1) exhibit consistency across all these four additional experiments. These quantitative measurements further highlight the influence of last-token late MLP modules on the prediction of r. We provide in Appendix J the heatmap illustrations of the effects for these additional studies.

5 Causal Effects on Different Tasks

In order to understand whether the patterns in the effect of the model components that we observed so far are specific to arithmetic queries, we compare our observations on arithmetic queries to two different tasks: the retrieval of a number from the prompt (§5.1), and the prediction of factual knowledge (§5.2). With this additional set of experiments, we aim to answer the question:

Q2 Are the activation patterns observed so far *specific* to the arithmetic setting?

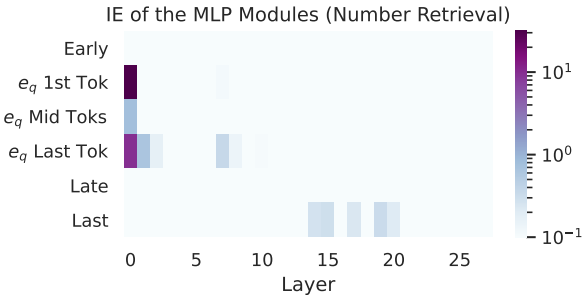

Figure 5: Indirect effect measured on the MLPs of GPT-J for predictions on the number retrieval task.

5.1 Information Flow on Number Retrieval

We consider a simple synthetic task involving numerical predictions. We construct a set of templates of the form "*Paul has* n_1 e_1 *and* n_2 e_2. *How many* e_q *does Paul have?*", where n_1, n_2 are two randomly sampled numbers, e_1 and e_2 are two entity names sampled at random,[6] and $e_q \in \{e_1, e_2\}$. In this case, the two input prompts p_1 and p_2 differ solely in the value of e_q. To provide the correct answer to a query, the model has simply to *retrieve* the correct number from the prompt. With this task, we aim to analyze the model's behavior in a setting involving numerical predictions but not requiring any kind of arithmetic computation.

We report the indirect effect measured for the MLPs modules of GPT-J in Figure 5. In this setting, we observe an unsurprising high-effect activation site corresponding to the tokens of the entity e_q and a lower-effect site at the end of the input in layers 14–20. The latter site appears in the set of the model components that were shown to be active on arithmetic queries. However, computing the relative importance of the late MLPs on this task shows that this second activation site is responsible for only $\mathrm{RI}(\mathcal{M}^{\text{late}}_{-1}) = 8.7\%$ of the overall $\log \mathrm{IE}$. The low RI, compared to the higher values measured on arithmetic queries, suggests that the function of the last-token late MLPs is not dictated by the numerical type of prediction, but rather by their involvement in *processing* the input information. This finding is aligned with our theory that

[5]The LLaMA tokenizer considers each digit as an independent token in the vocabulary. This makes it problematic to compare the probability value assigned by the model to multi-digit numbers. Therefore, we restrict the set of possible results to the set of single-digit numbers.

[6]We sample entities from a list containing names of animals, fruits, office tools, and other everyday items and objects.

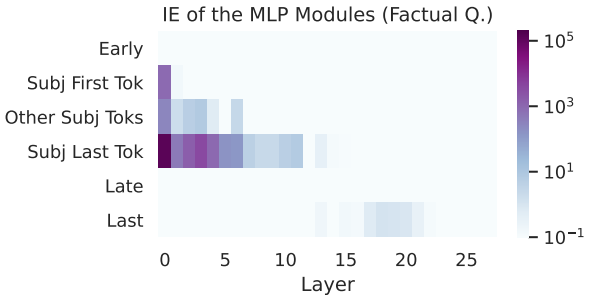

Figure 6: Indirect effect measured on the MLPs of GPT-J for predictions to factual queries.

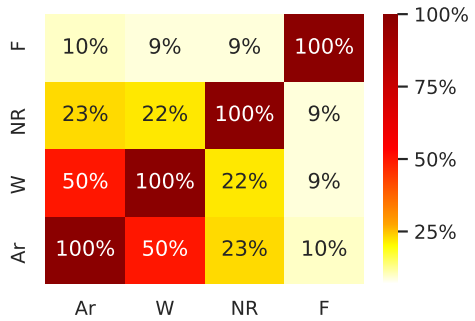

Figure 7: Overlap ratio in the top 400 neurons with the largest effect on predicting answers to factual queries involving Arabic Numerals (Ar) and numeral words (W), number retrieval (NR), and factual knowledge (F). The results are obtained with GPT-J.

sees $\mathcal{M}_{-1}^{\text{late}}$ as the location where information about r is included in the residual stream.

5.2 Information Flow on Factual Predictions

We carry out our experimental procedure using data from the LAMA benchmark (Petroni et al., 2019), which consists of natural language templates representing knowledge-base relations, such as "[subject] *is the capital of* [object]". By instantiating a template with a specific subject (e.g., *"Paris"*), we prompt the model to predict the correct object (*"France"*). Similar to our approach with arithmetic questions, we create pairs of factual queries that differ solely in the subject. In particular, we sample pairs of entities from the set of entities compatible for a given relation (e.g., cities for the relation "is the capital of"). Details about the data used for this procedure are provided in Appendix H. We then measure the indirect effect following the formulation in Equation 2, where the correct object corresponds to the correct numerical outcome in the arithmetic scenario.

From the results (Figure 6), we notice that a main activation site emerges in early layers at the tokens corresponding to the subject of the query. These findings are consistent with previous works (Meng et al., 2022; Geva et al., 2023), which hypothesize that language models store and retrieve factual associations in early MLPs located at the subject tokens. We compute the RI metric for the late MLP modules, which quantitatively validates the contribution of the early MLP activation site by attaining a low value of $\text{RI}(\mathcal{M}_{-1}^{\text{late}}) = 4.2\%$. The large IE observed at mid-sequence early MLPs represents a difference in the information flow with respect to the arithmetic scenario, where the modules with the highest influence on the model's prediction are located at the end of the sequence. This difference serves as additional evidence highlighting the specificity of the model's activation patterns when answering arithmetic queries.

5.3 Neuron-level Interventions

The experimental results in Sections 5.1 and 5.2 showed a quantitative difference in the contributions of last-token mid-late MLPs between arithmetic queries and two tasks that do not involve arithmetic computation. Now, we investigate whether the components active *within* $\mathcal{M}_{-1}^{\text{late}}$ on the different types of tasks are different. We carry out a finer-grained analysis in which we consider independently each neuron in an MLP module (i.e., each dimension in the output vector of the function $\text{MLP}^{(l)}$) at a specific layer l. In particular, following the same procedure as for layer-level experiments, we intervene on each neuron by setting its activation to the value it would take if the input query contained different operands (or a different entity). We then compute the corresponding indirect effect as in Eq. 2. We carry out this procedure for arithmetic queries using Arabic numerals (Ar) and numeral words (W), for the number retrieval task (NR), and for factual knowledge queries (F).[7] We rank the neurons according to the average effect measured for each of these four settings and compute the overlap in the top 400 neurons (roughly 10%, as GPT-J has a hidden dimension of 4096).

We carry out this procedure for layer $l = 19$, as it exhibits the largest IE within $\mathcal{M}_{-1}^{\text{late}}$ on all the tasks considered. The heatmap in Figure 7 illustrates the results. We observe a consistent overlap (50%) between the top neurons active for the arithmetic queries using Arabic and word-based representa-

[7] To have the same result space for all the arithmetic queries (Ar and NW) and for the number retrieval task, we restrict the set $\mathcal{S}$ to $\{1, \ldots, 20\}$ (or the corresponding numeral words).

tions. Interestingly, the size of the neuron overlap between arithmetic queries and number retrieval is considerably lower (22% and 23%), even though both tasks involve the prediction of numerical quantities. Finally, the overlaps between the top neurons for the arithmetic operations and the factual predictions (between 9% and 10%) are not larger than for random rankings: the expected overlap ratio between the top 400 indices in two random rankings of size 4096 is 9.8% (Antverg and Belinkov, 2022). These results support the hypothesis that the model's circuits responsible for different kinds of prediction, though possibly relying on similar subsets of layers, are distinct. However, it is important to note that this measurement does not take into account the magnitude of the effect.

6 Conclusion

We proposed the use of causal mediation analysis to mechanistically investigate how LMs process information related to arithmetic. Through controlled interventions on specific subsets of the model, we assessed the impact of these mediators on the model's predictions.

We posited that models produce predictions to arithmetic queries by conveying the math-relevant information from the mid-sequence early layers to the last token, where this information is then processed by late MLP modules. We carried out a causality-grounded experimental procedure on four different Transformer-based LMs, and we provided empirical evidence supporting our hypothesis. Furthermore, we showed that the information flow we observed in our experiments is specific to arithmetic queries, compared to two other tasks that do not involve arithmetic computation.

Our findings suggest potential avenues for research into model pruning and more targeted training/fine-tuning by concentrating on specific model components associated with certain queries or computations. Moreover, our results offer insights that may guide further studies into using LMs' hidden representations to correct the model's behavior on math-based tasks at inference time (Li et al., 2023) and to estimate the probability of the model's predictions to be true (Burns et al., 2023).

Limitations

The scope of our work is investigating arithmetic reasoning and we experiment with the four fundamental arithmetic operators. Addition, subtraction, multiplication, and division form the cornerstone of arithmetic calculations and serve as the basis for a wide range of mathematical computations. Thus, exploring their mechanisms in language models provides a starting point to explore more complex forms of mathematical processing. Studying a broader set of mathematical operators represents an interesting avenue for further investigation.

Our work focuses on synthetically-generated queries that are derived from natural language descriptions of the four basic arithmetic operators. To broaden the scope, future research can expand the analysis of model activations to encompass math-based queries described in real-life settings, such as math word problems.

Finally, a limitation of our work concerns the analysis of different attention heads. In our experiments, we consider the output of an attention module as a whole. Future research could focus on identifying the specific heads that are responsible for forwarding particular types of information in order to offer a more detailed understanding of their individual contributions.

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